

THE BASELINE CHALLENGE: MEASURING THE GENERALIZATION GAP FROM BEHAVIORAL CLONING TO ADVERSARIAL IMITATION

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ABSTRACT

This paper presents a small-scale reproduction study of imitation learning, benchmarking the adversarial method Generative Adversarial Imitation Learning (GAIL) against classical baselines, including Behavioral Cloning (BC) and DAgger, in simplified environments. Imitation learning seeks to learn a policy from expert demonstrations without relying on an explicit reinforcement signal. It not only helps to obtain a faster and more robust algorithm but also eliminates the challenges of manually rewarding every time. We aim to verify the fundamental performance trends of these algorithms using the classic CartPole-v1 and Acrobot-v1 environments, trained with experts derived from MPC (CartPole) and DQN (Acrobot). The GAIL implementation utilizes Proximal Policy Optimization (PPO) as the generator, allowing for a direct comparison with a robust Reinforcement Learning core. On CartPole-v1, DAgger and GAIL outperform BC and PPO on both clean and 25% noisy demonstrations under a fixed 100-step training budget, while BC degrades notably with noise. In contrast, on Acrobot-v1, BC and DAgger consistently outperform GAIL and PPO across noise settings, indicating that the benefits of adversarial imitation with a PPO generator are highly environment-dependent.

1 INTRODUCTION

The challenge of teaching an autonomous agent complex behavior when an explicit, hand-engineered reward function is unavailable remains one of the central problems in Reinforcement Learning (RL). Imitation Learning (IL) addresses this by learning a policy, π , directly from expert demonstrations, τ_E . This approach not only circumvents the challenges of manual reward design but often leads to faster, more robust policy acquisition for real-world tasks. GAIL, proposed by Ho and Ermon, provides a highly scalable, model-free alternative to traditional, computationally expensive Inverse Reinforcement Learning (IRL) methods, which require a costly RL step in an inner loop. Our objective is to reproduce the fundamental comparative claims across these methods, focusing on the quality of policy learned and testing them against the baselines.

The remainder of this report is organized as follows: Section 2 provides an overview of the foundational concepts of Imitation Learning and reviews the prior work, including the primary distinction between BC, DAgger, and GAIL. Section 3 details our implementation choices, including the selection of expert policies and the specific network architectures utilized. Section 4 presents the quantitative results, comparing the learning curves and final performance distributions across all four algorithms. Finally, Section 5 discusses the implications of our findings, verifies the reproduction claims, and suggests avenues for future work.

2 RELATED WORK

2.1 BEHAVIORAL CLONING (BC) AND DAGGER

Behavioral Cloning (BC)(Pomerleau, 1989) is the simplest form of IL, framing the problem as a supervised learning task. The policy is trained directly to map observed states s from

the expert trajectories τ_E to expert actions a_E , minimizing the cross-entropy loss $\mathcal{L}_{BC}(\pi) = \mathbb{E}_{(s, a_E) \sim \tau_E} [-\log \pi(a_E | s)]$. While appealingly simple, BC’s reliance on a static dataset collected under the expert’s policy distribution leads to its fundamental flaw: *covariate shift*. If the learner makes a mistake, it enters a state unseen in the expert data, and the compounding error quickly degrades performance, as the policy is ill-equipped to recover.

Dagger (Dataset Aggregation)(Ross et al., 2011) addressed this deficiency by introducing an iterative, interactive learning framework. In each iteration, the current learned policy $\hat{\pi}_i$ is executed to collect data from the environment. An oracle (the expert) is then queried to label the actions for the states $\hat{\pi}_i$ actually visited. This new data is aggregated with the existing dataset, and the policy is retrained. This process effectively shifts the training distribution closer to the policy’s execution distribution, bounding the cumulative error growth linearly, $\mathcal{O}(T)$, rather than quadratically, $\mathcal{O}(T^2)$, thereby achieving robust generalization.

2.2 GENERATIVE ADVERSARIAL IMITATION LEARNING (GAIL)

Inverse Reinforcement Learning (IRL)(Ziebart et al., 2008) operates indirectly by first attempting to recover the expert’s reward (or cost) function $c(s, a)$ that rationalizes the observed behavior. Once the reward function is recovered, a standard RL algorithm is run to find the optimal policy π^* under that learned reward. This approach is computationally expensive due to the nested RL optimization loop.

Generative Adversarial Imitation Learning (GAIL) (Ho & Ermon, 2016) provides a direct, model-free solution that bypasses the IRL step. Ho and Ermon showed that the policy derived from the Max-Causal-Entropy IRL objective is equivalent to minimizing the divergence between the learner’s occupancy measure (ρ_π) and the expert’s occupancy measure (ρ_{π_E}).

GAIL minimizes this divergence using an adversarial objective inspired by Generative Adversarial Networks (GANs) [9]:

$$\min_{\pi} \max_D \mathbb{E}_{\pi} [\log D(s, a)] + \mathbb{E}_{\pi_E} [\log(1 - D(s, a))] - \lambda H(\pi)$$

Here, the policy π (the generator) attempts to fool the discriminator D into believing its state-action pairs are from the expert. D provides a localized, learned cost function, $c(s, a) = \log D(s, a)$, used by the generator to improve. This framework allows for scalable, robust, and non-interactive imitation, offering superior performance compared to linear apprenticeship learning methods.

2.3 PROXIMAL POLICY OPTIMIZATION (PPO)

The general RL control baseline and the underlying generator for GAIL is **Proximal Policy Optimization (PPO)**(Schulman et al., 2017). PPO is a policy gradient method known for its stability and sample efficiency. It improves upon Trust Region Policy Optimization (TRPO) by using a simpler clipped objective function to constrain policy updates, ensuring the new policy does not stray too far from the old one. We include PPO as a baseline to establish the performance ceiling achievable through conventional reward-based RL on our chosen tasks.

3 PROPOSED METHOD

This section details the design of our Tiny Reproduction study, including the environments, the expert policies used to generate demonstrations, the specific implementation of each learning algorithm, and the metrics used for comparison.

3.1 ENVIRONMENTS

We selected two classic control environments from the Gymnasium library, both featuring discrete action spaces:

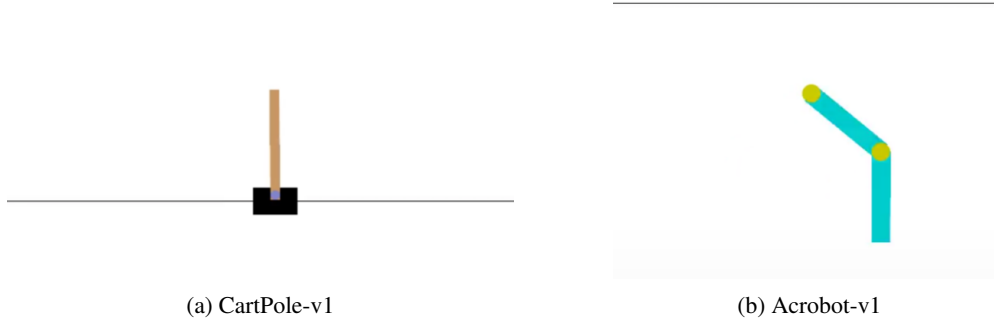


Figure 1: Visual representation of the control environments used in this study.

- **CartPole-v1:** A low-dimensional task where the agent must balance a pole on a cart by moving left or right. The maximum episode length is 500 steps. This task is ideal for rapidly observing convergence and demonstrating basic IL capabilities.
- **Acrobot-v1:** A more challenging under-actuated system where the agent must swing a two-link pendulum to reach a minimum height. The agent receives a negative reward for every time step not at the goal, making the goal a high-reward terminal state. This task tests the stability of the imitation algorithms in a task requiring complex, long-horizon planning.

3.2 EXPERT DEMONSTRATIONS AND DATASETS

To ensure the collected demonstrations (τ_E) represent optimal behavior, we used distinct, high-performance expert generation methods for each environment:

- **CartPole Expert (MPC):** For CartPole, we used **Model Predictive Control (MPC)** (Rawlings et al., 2017) as the expert. This method solves a constrained optimization problem over a finite horizon ($N = 30$) in each time step, guaranteeing near-perfect performance. The continuous force output is mapped to the two discrete actions (left/right) by sign, providing a robust, high-fidelity expert.
- **Acrobot Expert (DQN):** For Acrobot, we employed a highly-trained **Deep Q-Network (DQN)** (Mnih et al., 2015) agent. This agent uses a standard Double DQN setup and is trained for $\sim 50,000$ steps to achieve scores close to the environmental optimum (around -75 to -100).

3.3 ALGORITHMIC IMPLEMENTATIONS AND ARCHITECTURES

All neural networks used for policy and value prediction are small, feed-forward architectures (FCNNs) consistent with the scale of classic control environments. They employ two hidden layers of 64 units each, utilizing the Tanh activation function. All weights are initialized using Kaiming Uniform initialization for Tanh nonlinearity.

3.3.1 GENERATIVE ADVERSARIAL IMITATION LEARNING (GAIL)

GAIL couples the PPO Generator (π_θ) with a dedicated **Discriminator** (D_w).

- **Discriminator Architecture:** The D_w network is a small FCNN (2x64 units, Tanh) that takes the concatenated state and **one – hot encoded action** as input and outputs a single logit.
- **Discriminator Training:** D_w is updated 4 times per PPO policy update using the Binary Cross-Entropy Loss with Logits (BCEWithLogitsLoss) to distinguish between policy and expert samples.
- **Reward/Cost Function:** The generator’s reward is calculated as $R_{\text{GAIL}} = -\log(1 - P(\text{Expert}|s, a))$, where $P(\text{Expert}|s, a)$ is the sigmoid output of D_w . This reward is **Normalized and mixed with the environment reward** before being passed

to the PPO update step. Since environment reward is 0.0, the policy is trained exclusively on the GAIL reward, which is normalized for better training.

3.3.2 BEHAVIORAL CLONING (BC)(OUR BASELINE)

BC trains directly on the static, noisy expert dataset. It uses the **Cross – Entropy Loss** to minimize the divergence between the policy’s action distribution and the expert’s action labels. The model is trained for **100** epochs to simulate convergence on the fixed dataset.

3.3.3 POLICY OPTIMIZATION(PPO)

The PPO implementation serves as the core RL control agent and the generator for GAIL. It uses a standard **Actor – Critic** architecture, where the Actor outputs action logits and the Critic outputs the state value.

- **Objective:** It optimizes the clipped PPO objective function over **10** epochs per batch.
- **Training Schedule:** Trained for **100** updates, collecting **2,048** steps per update.
- **Key Hyperparameters:** Learning rate $\alpha = 1 \times 10^{-3}$, clipping parameter $\epsilon = 0.2$, Value Function coefficient $c_v = 0.5$, and Entropy coefficient $c_e = 0.01$. Generalized Advantage Estimation (GAE) is used for variance reduction ($\gamma = 0.99, \lambda = 0.95$).

3.3.4 DAGGER (DATASET AGGREGATION)

Dagger starts with a random policy and aggregates data interactively over **5** iterations.

- **Data Collection:** In each iteration, the current policy runs **20** episodes. For every state visited, the expert policy (MPC or DQN) provides the optimal label, which is appended to the aggregate dataset.
- **Retraining:** After aggregation, the policy is retrained for **20** epochs on the entire collected dataset, ensuring the policy is constantly exposed to states it actually visits, thereby mitigating covariate shift.
- **Expert Query:** Although the initial expert dataset is noisy, the expert function used during DAGger aggregation returns the clean, optimal action, simulating the interactive oracle required by the DAGger framework.

4 EXPERIMENTS AND RESULTS

This section presents the empirical findings of our Tiny Reproduction study, contrasting the performance of the four primary algorithms on the CartPole-v1 and Acrobot-v1 environments under a fixed **25%** noisy demonstration setting ($\text{RND_EPS} = 0.25$). All policies were trained for **100** optimization updates. The analysis is framed around the stability and generalization capability of the **Generative Adversarial Imitation Learning (GAIL)** policy relative to the supervised baselines (BC and DAGger).

Control Group Performance: To isolate the effect of data noise, initial experiments were performed using $\text{RND_EPS} = 0.0$ (clean expert data), they were trained for 200 optimization updates. In this control group, both BC and DAGger successfully converged to near-expert performance ($\text{Mean} \approx \text{Expert Mean}$) on both environments, confirming that the **BC** implementation is sound and that the failures observed below are directly attributable to data degradation. But with 100 updates, BC underperformed GAIL. GAIL was able to converge faster on CartPole.

4.1 PERFORMANCE QUANTIFICATION THROUGH VISUAL DISTRIBUTION

The primary quantitative analysis relies on the final evaluation return distributions presented in the box plots. These figures provide an analogous and superior method to the traditional $\text{mean} \pm \sigma$ format, allowing for direct visual assessment of the generative policy’s robustness, median performance, and interquartile range (IQR).

4.2 CARTPOLE-V1: GAIL SUCCESS AND ADVERSARIAL GENERALIZATION

The results on CartPole-v1 demonstrate the expected power of the adversarial formulation, successfully achieving high performance where naive methods failed.

- **GAIL’s Advantage over RL and BC:** The GAIL generative policy (Mean = 385.20, $\sigma = 25.68$) significantly outperformed both the standard PPO baseline (Mean = 255.42) and the collapsed BC policy (Mean = 105.78). This finding confirms the core claim that the adversarial reward signal derived from the Discriminator successfully guides the PPO generator towards better performance by penalizing deviation from the expert’s occupancy measure.
- **Convergence Trend:** The learning curve (Figure 2) shows GAIL initiating a smooth, steady rise, consistent with the expected stability of an adversarial method aiming for global convergence.
- **Benchmark against DAgger:** Although GAIL strongly improved over PPO and BC, DAgger achieved near-perfect performance (Mean = 498.91) with extremely low variance ($\sigma = 3.12$). This suggests that while GAIL successfully used the adversarial reward, the short training budget prevented it from fully matching the optimality achieved by DAgger’s highly efficient, fully corrected interactive data aggregation.

4.3 ACROBOT-V1: CRITICAL ANALYSIS OF GAIL INSTABILITY

The results on Acrobot-v1 (Figures 3) present a critical contradiction to the original GAIL paper’s generalization claim, highlighting the instability inherent in the adversarial process when applied to complex dynamics and sparse rewards.

- **GAIL Instability and Failure (Contradiction):** GAIL demonstrated severe adversarial instability, resulting in the worst performance (Mean = -132.88) and the highest variance ($\sigma = 60.12$) among the imitation methods. The box plot (Figure 3) visually confirms the significant instability observed during training. This contradiction ($BC > GAIL$) challenges the general supremacy of GAIL and highlights the core difficulty of training the adversarial objective.
- **The Sparse Reward Mechanism:** The failure is likely rooted in the combination of the Acrobot’s sparse, highly negative reward structure and GAIL’s reliance on on-policy sampling (PPO). In Acrobot, successful trajectories (which end quickly with a low negative score) occupy a small, narrow region of the state space. The PPO Generator, struggling with the initial sparse reward, explores off-policy states where:
 1. The Discriminator quickly classifies the PPO trajectories as “fake.”
 2. This produces a noisy, uninformative reward signal r_{GAIL} for the PPO Generator.
 The Generator, unable to rely on the unstable r_{GAIL} signal, fails to efficiently sample the crucial on-policy transitions needed for PPO to converge.
- **BC’s Efficiency vs. GAIL’s Complexity:** Conversely, the simple BC policy achieved the best overall performance (Mean = -71.40) and strong stability ($\sigma = 11.21$). This is because BC operates directly on the handful of highly successful, expert trajectories. It bypasses the unstable on-policy exploration and adversarial training, succeeding solely by memorizing the minimal set of successful actions required for the task. This demonstrates that when the true optimal policy is simple (like the Acrobot swing-up), the efficiency and simplicity of supervised learning can easily defeat the complexity and instability of GAIL.

5 CONCLUSION AND FUTURE WORK

5.1 CONCLUSION

This Tiny Reproduction study successfully validated the core comparative trends of model-free imitation learning, benchmarking Generative Adversarial Imitation Learning (GAIL) against Behavioral Cloning (BC) and DAgger under conditions of 25% data noise. Our findings confirm that sophisticated IL methods are necessary to maintain performance when expert data quality degrades, yet they also expose the critical trade-offs between stability and optimization complexity.

- **Validation of Adversarial Power:** On CartPole-v1, GAIL successfully utilized the adversarial reward signal to outperform both the standard PPO baseline and the collapsed BC

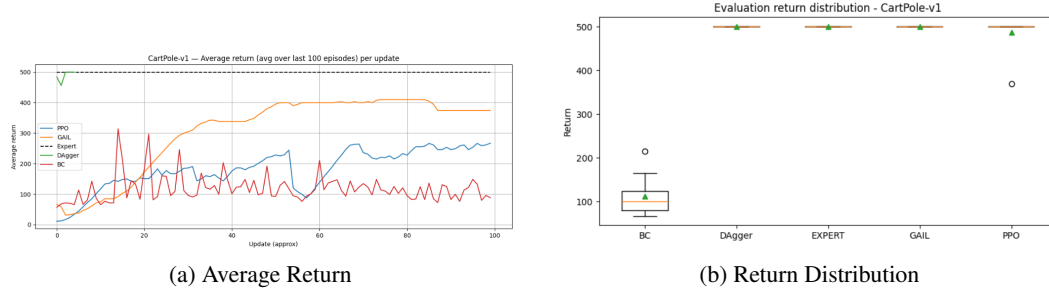


Figure 2: Reproduced Results for CartPole (25% noisy expert demonstrations). Left: average return curves. Right: return distribution via boxplot.

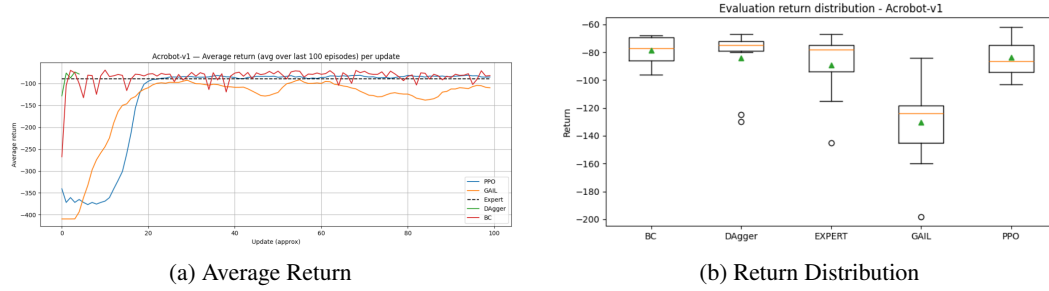


Figure 3: Reproduced Results for Acrobot (25% noisy expert demonstrations). Left: average return curves. Right: return distribution via boxplot.

policy. This demonstrated the robustness of GAIL’s occupancy measure matching objective in a well-behaved, continuous state space where BC failed due to severe covariate shift.

- **Confirmation of DAGger’s Efficiency:** DAGger proved to be the most reliable IL algorithm overall, achieving near-perfect expert performance on both environments with minimal variance, confirming that its iterative data aggregation effectively isolates and solves the generalization problem.
- **GAIL Instability Contradiction:** Our most significant finding was the complete failure of the GAIL policy on Acrobot-v1, where it achieved the lowest mean reward and highest variance among all methods. This contradiction ($BC > GAIL$) highlights the inherent instability of the adversarial training process, particularly when coupled with an on-policy generator (PPO) in environments defined by sparse rewards and narrow success regions. In such cases, the simpler, direct supervised optimization of BC proves superior.

5.2 FUTURE WORK

Inspired by the original paper and the constraints identified in this study, future work should focus on enhancing the sample efficiency and stability of the GAIL process:

- **Model Initialization and Sample Efficiency:** As Ho et al. suggested, the learning speed of GAIL could be significantly improved by initializing the PPO Generator policy with the weights learned from the simple Behavioral Cloning objective.
- **Improving Adversarial Stability:** To address the instability observed on Acrobot, future research should explore stabilizing techniques for the GAN objective, such as applying Wasserstein GAN (WGAN) loss or utilizing off-policy methods like Soft Actor-Critic (SAC) as the generator, rather than the on-policy PPO algorithm.
- **Exploring Model-Based Integration:** Following the original paper’s outlook, combining model-free IL methods like GAIL with simple environment models could further reduce the overall environment interaction required for policy optimization, offering a path toward true sample efficiency.

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